

Análisis del Impacto de la Inteligencia Artificial

Impacto en el Desarrollo de Software

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06 de June de 2026

Resumen

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequale doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et.

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1. Introducción

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequaleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet,

ut et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum.

2. Estado del Arte

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequaleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet, ut et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum defuturum, quas natura non depravata desiderat. Et quem ad me accedis, saluto: 'chaere,' inquam, 'Tite!' lictores, turma omnis chorusque: 'chaere, Tite!' hinc hostis mi Albucius, hinc inimicus. Sed iure Mucius.

3. Metodología

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequaleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet, ut et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum defuturum, quas natura non depravata desiderat. Et quem ad me accedis, saluto: 'chaere,' inquam, 'Tite!' lictores, turma omnis chorusque: 'chaere, Tite!' hinc hostis mi Albucius, hinc inimicus. Sed iure Mucius. Ego autem

mirari satis non queo unde hoc sit tam insolens domesticarum rerum fastidium. Non est omnino hic docendi locus; sed ita prorsus existimo, neque eum Torquatum, qui hoc primus.

4. Página 2

5. Introducción

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequale doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet, ut et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum defuturum, quas natura non depravata desiderat. Et quem ad me accedis, saluto: 'chaere,' inquam, 'Tite!' lictores, turma omnis chorusque: 'chaere, Tite!' hinc hostis mi Albucius, hinc inimicus. Sed iure Mucius. Ego autem mirari satis non queo unde hoc sit tam insolens domesticarum rerum fastidium. Non est omnino hic docendi locus; sed ita prorsus existimo, neque eum Torquatum, qui hoc primus cognomen invenerit, aut torquem illum hosti detraxisse, ut aliquam ex eo est consecutus? – Laudem et caritatem, quae sunt vitae sine metu degendae praesidia firmissima. – Filium morte multavit. – Si sine causa, nollem me ab eo delectari, quod ista Platonis, Aristoteli, Theophrasti orationis ornamenta neglexerit. Nam illud quidem physici, credere aliquid esse minimum, quod profecto numquam putavisset, si a Polyano, familiari suo, geometrica discere maluisset quam illum etiam ipsum.

6. Metodologías

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequale

doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet, ut et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum.

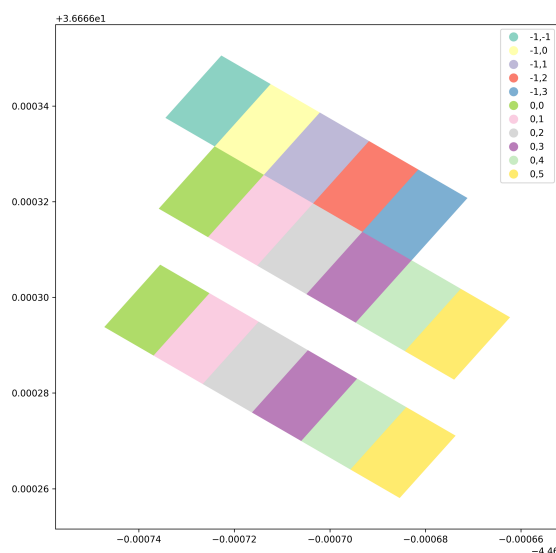


Figura 1: Descripción de los paneles

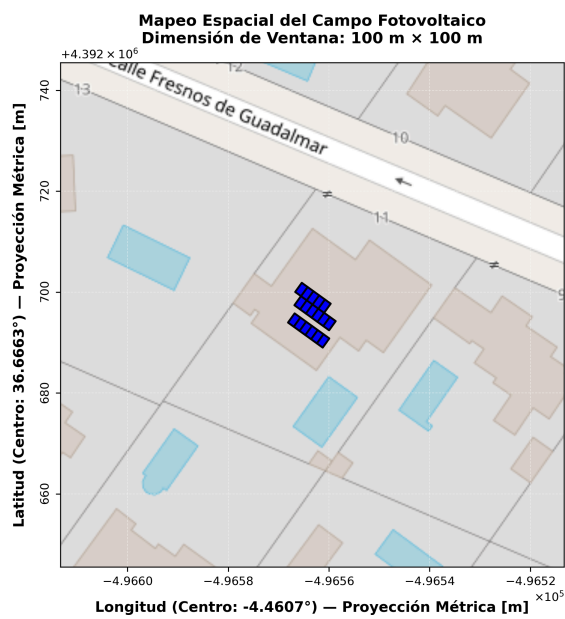


Figura 2: Descripción de los paneles

7. df calculados

lat	lon	peakpower	loss	fixed	usehorizon	userhorizon	pvprice	lifetime	interest	angle	inclinacion	systemcost	outputformat	grupo
36.66629732748714	−4.460736215114594	540	14	1	1	1,1,40,45,50,55,1,1,1,1,1,1,1,1,1	1	20	5	0	30	8	json	0
36.66632206941558	−4.460724815726281	540	14	1	1	1,1,1,1,1,1,1,1,1,1,1,1,1,1,1	1	20	5	0	30	8	json	1

Tabla 1: pvigis params

8. inputs

Categoría	Variable	valor
Proyecto	nombre	Mi rrrr Real
Autor		Pepe Perez
	N Registro	N.12345
Organización	nombre	Instalaciones S.A.
	CIF	A12345678
	dirección	Calle Iglesia, 123
	teléfono	912 34 56 78
	correo	pperezinstalaciones.com
Destinatario	nombre	Ayto. Villanueva
	CIF	
	dirección	Plaza Mayor, 1

Tabla 2: PDF

Ca- pí- tulo	Ca- te- go- ría	nom- bre	tipoEs- truc- tu- ra	id	lat	lng	ro- ta- tion	width	height	tilt	slo- pe	orien- ta- tion	mo- de- lo	po- tenciaW	rows	cols	in- ver- ter- Mo- del	in- ver- ter- Po- wer	mppt- Count	vOc	vMp	iSc	iMp	coef- Voc	vMax	vMinMp	ppMaxMp	iMa- xIn- ver- ter	iSc- MaxIn- ver- ter	pa- ne- les	som- bras
FV	Pa- ne- les	G1c	co- pla- nar	17796183948729732508352750175947892663	39.48729732508352750175947892663	-89.2663	30	0	ver- ti- cal	JAM60D42- LB	540	1	1	De- ye SUN-5K- SG05 5000 LP1- EU-AM2- P	2	44.33	37.25	15.36	14.5	0.25	500	150	425	18	27	[“0,0”,“0,1”,“0,2”,“0,3”,“0,4”,“0,5”,“0,6”,“0,7”,“0,8”,“0,9”,“1,0”,“1,1”,“1,2”,“1,3”,“1,4”,“1,5”,“1,6”,“1,7”,“1,8”,“1,9”,“2,0”,“2,1”,“2,2”,“2,3”,“2,4”,“2,5”,“2,6”,“2,7”,“2,8”,“2,9”,“3,0”,“3,1”,“3,2”,“3,3”,“3,4”,“3,5”,“3,6”,“3,7”,“3,8”,“3,9”,“4,0”,“4,1”,“4,2”,“4,3”,“4,4”,“4,5”,“4,6”,“4,7”,“4,8”,“4,9”,“5,0”,“5,1”,“5,2”,“5,3”,“5,4”,“5,5”,“5,6”,“5,7”,“5,8”,“5,9”,“6,0”,“6,1”,“6,2”,“6,3”,“6,4”,“6,5”,“6,6”,“6,7”,“6,8”,“6,9”,“7,0”,“7,1”,“7,2”,“7,3”,“7,4”,“7,5”,“7,6”,“7,7”,“7,8”,“7,9”,“8,0”,“8,1”,“8,2”,“8,3”,“8,4”,“8,5”,“8,6”,“8,7”,“8,8”,“8,9”,“9,0”,“9,1”,“9,2”,“9,3”,“9,4”,“9,5”,“9,6”,“9,7”,“9,8”,“9,9”]					
		G1	co- pla- nar	17796183948729732508352750175947892663	39.48729732508352750175947892663	-89.2663	30	0	ver- ti- cal	JAM60D42- LB	540	1	1	De- ye SUN-5K- SG05 5000 LP1- EU-AM2- P	2	44.33	37.25	15.36	14.5	0.25	500	150	425	18	27	[“0,0”,“0,1”,“0,2”,“0,3”,“0,4”,“0,5”,“0,6”,“0,7”,“0,8”,“0,9”,“1,0”,“1,1”,“1,2”,“1,3”,“1,4”,“1,5”,“1,6”,“1,7”,“1,8”,“1,9”,“2,0”,“2,1”,“2,2”,“2,3”,“2,4”,“2,5”,“2,6”,“2,7”,“2,8”,“2,9”,“3,0”,“3,1”,“3,2”,“3,3”,“3,4”,“3,5”,“3,6”,“3,7”,“3,8”,“3,9”,“4,0”,“4,1”,“4,2”,“4,3”,“4,4”,“4,5”,“4,6”,“4,7”,“4,8”,“4,9”,“5,0”,“5,1”,“5,2”,“5,3”,“5,4”,“5,5”,“5,6”,“5,7”,“5,8”,“5,9”,“6,0”,“6,1”,“6,2”,“6,3”,“6,4”,“6,5”,“6,6”,“6,7”,“6,8”,“6,9”,“7,0”,“7,1”,“7,2”,“7,3”,“7,4”,“7,5”,“7,6”,“7,7”,“7,8”,“7,9”,“8,0”,“8,1”,“8,2”,“8,3”,“8,4”,“8,5”,“8,6”,“8,7”,“8,8”,“8,9”,“9,0”,“9,1”,“9,2”,“9,3”,“9,4”,“9,5”,“9,6”,“9,7”,“9,8”,“9,9”]					

Tabla 3: Diseño FV

Capítulo	Categoría	Subcategoría	Consumo_diario_kWh	Perfil_24h
Residencial	Vivienda Estándar	Uso General	10	[0.024, 0.016, 0.016, 0.016, 0.016, 0.024, 0.032, 0.145, 0.04, 0.032, 0.024, 0.024, 0.032, 0.04, 0.032, 0.024, 0.032, 0.047, 0.063, 0.08, 0.095, 0.071, 0.047, 0.032]
	Vivienda Unifamiliar	Aire Acondicionado	12	[0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.02, 0.05, 0.08, 0.12, 0.15, 0.15, 0.12, 0.1, 0.08, 0.06, 0.04, 0.02, 0.01, 0.0, 0.0, 0.0]
		Calefacción (B. Calor)	20	[0.05, 0.06, 0.07, 0.07, 0.07, 0.08, 0.1, 0.12, 0.08, 0.05, 0.03, 0.02, 0.02, 0.02, 0.02, 0.02, 0.02, 0.03, 0.04, 0.05, 0.06, 0.06, 0.04, 0.04]
		Cargador Coche Eléctrico	15	[0.12, 0.12, 0.12, 0.12, 0.12, 0.12, 0.12, 0.12, 0.04, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.04]
	Bloque Pisos	Servicios Comunes	30	[0.0416, 0.0416]
Terciario	Oficina Moderna	Iluminación LED	15	[0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.05, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.05, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0]
		Climatización	50	[0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.05, 0.1, 0.12, 0.12, 0.12, 0.12, 0.05, 0.12, 0.12, 0.06, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0]
	Comercio	Rótulos Luminosos	5	[0.1, 0.1, 0.1, 0.1, 0.1, 0.05, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.05, 0.1, 0.1, 0.1, 0.1]
Industrial	Nave Producción	Maquinaria Pesada	350	[0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.05, 0.15, 0.15, 0.15, 0.15, 0.15, 0.05, 0.15, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0]
		Aire Comprimido	40	[0.0416, 0.0416]
	Almacén Logístico	Carga Carretillas	60	[0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.1, 0.1, 0.1, 0.1]
Agrario	Explotación Agrícola	Bombeo Riego	150	[0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.1, 0.15, 0.15, 0.15, 0.15, 0.15, 0.15, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0]
	Granja Ganadera	Ventilación Forzada	25	[0.0416, 0.0416]
		Ordeño Automático	15	[0.0, 0.0, 0.0, 0.0, 0.0, 0.2, 0.2, 0.1, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.2, 0.2, 0.1, 0.0, 0.0, 0.0, 0.0]

Tabla 4: CONSUMO

Capítulo	Categoría	Variable	Fijo (€)	€/Wdc	MO (%)
Coste Directo	Equipos Principales	Paneles fotovoltaicos monocristalinos	0	0.22	5
		Inversor string/híbrido	150	0.12	5
		Sistema de almacenamiento (Batería)	3000	0.15	5
	Balance de Sistema (BOS)	Soportes, cableado y protecciones	750	0.19	40
		Sistema de monitorización	150	0.02	10
	Obra Civil	Adecuación y pequeñas adaptaciones	200	0.005	80
Coste Indirecto	Ingeniería	Proyecto, dirección y legalización	900	0.02	100
	Interconexión	Trámites con distribuidora y tasas	500	0.01	90
	Gastos Generales	Margen comercial y overhead	0	0.18	0
	Contingencias	Reserva para imprevistos (3-5%)	0	0.04	0
Otros	Otros		0	0.0	0

Tabla 5: COSTES DE INSTALACION

Capítulo	Categoría	Concepto	Año_inicio	Año_fin	Fijo (€/año)	Variable	Unidad_variable
Operación FV	Energía fotovoltaica	PV_Fijo_Anual	1	25	0.0	0	€/año
		PV_Fijo_por_Capacidad	1	25	19.0	0	€/Wdc
		PV_Variable_Generacion	1	25	0.0	0	€/kWh
Batería	Almacenamiento	Bateria_Fijo_Anual	1	15	0.0	0	€/año
		Bateria_Fijo_por_Capacidad	1	15	7.25	0	€/kWh_instalada
		Bateria_Variable_Descarga	1	15	0.0	0	€/kWh_descargado
	Reemplazo de activos	Bateria_Coste_Reemplazo	10	10	252.0	0	€/kWh_instalada
Infraestructura	Uso de suelo	Arrendamiento_Anual_Terreno	1	25	0.0	0	€/año
Otrsos	Otdros	Otros	1	25	0.0	0	€

Tabla 6: COSTES DE OPERACION

Capítulo	Categoría	Concepto	Año_inicio	Año_fin	Ingreso_fijo (€/año)	Ingreso_variable	Unidad_variable
Ingresos de explotación	Autoconsumo	Ahorro autoconsumo	1	25	0	0.18	€/kWh autoconsumido
	Excedentes	Venta excedentes	1	25	0	0.08	€/kWh exportado
		Compensación simplificada	1	25	0	0.06	€/kWh compensado
Ingresos regulados	Certificados	CAE	1	10	0	0.14	€/kWh generado
		Garantías de origen	1	25	0	0.005	€/kWh generado
Subvenciones	Bonificaciones fiscales	Bonificación IBI	1	5	100	0.0	€/año
	Deducciones fiscales	Deducción IRPF	1	1	200	0.0	€/año
	Inversión	Next Generation	1	1	0	0.3	% CAPEX
	Locales/autonómicas	ICIO	1	1	100	0.0	€/año
Otros	Otros	Otros	1	25	0	0.0	€

Tabla 7: INGRESOS

Capítulo	Categoría	Parámetro	Valor	Unidad
Análisis Financiero	Proyecto	Duración de la actuación	2033.0	años
		Impuestos (IVA)	21.0	%
	Mercado	Tasa de inflación	2.5	%/año
		Precio de la energía	0.12	€/kWh
		Tasa de descuento (VAN)	0.08	-
	Financiación	Porcentaje financiado	27.0	%
		Años de financiación	10.0	años
		Tipo de interés anual	7.0	%
	Operación	Tasa seguro anual	0.0	% de instalación

Tabla 8: FINANCIEROS

8.1. Llave: Diseño_FV_gdf

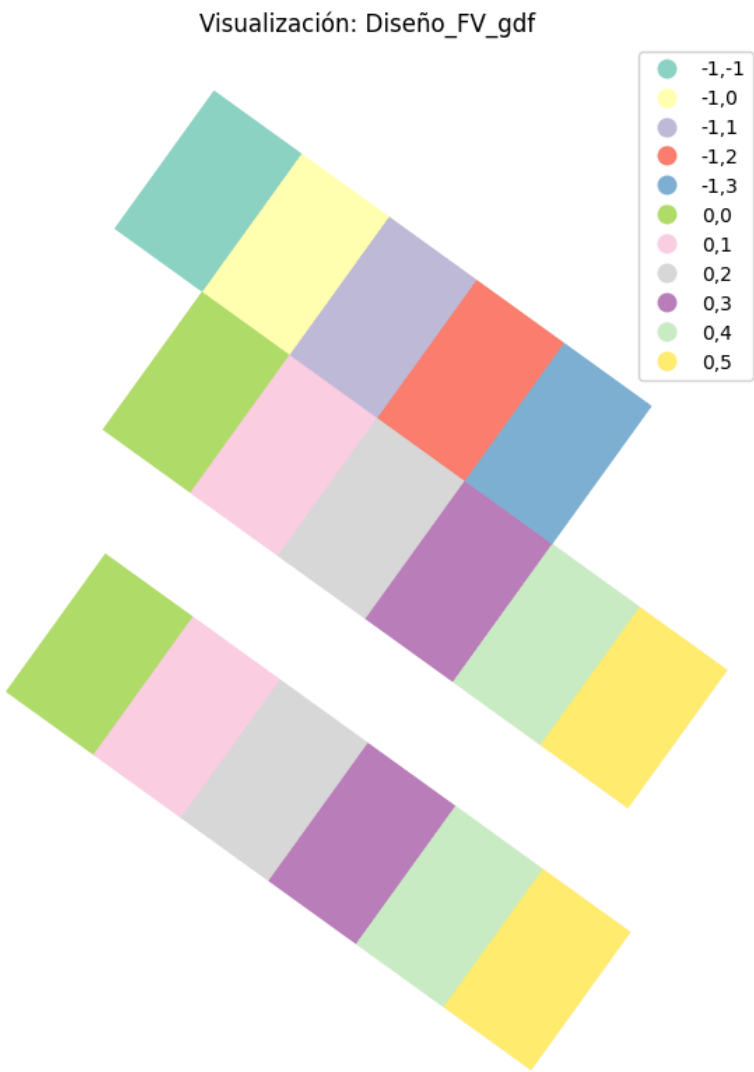


Figura 3: Descripción de los paneles

9. jsonsdd

9.1. G1c

- **inputs:**
 - **location:**
 - **latitude:** 36.66629732748714
 - **longitude:** −4.460736215114594
 - **elevation:** 3.0
 - **meteo_data:**
 - **radiation_db:** PVGIS-SARAH3
 - **meteo_db:** ERA5
 - **year_min:** 2005
 - **year_max:** 2023
 - **use_horizon:** True
 - **horizon_db:** DEM-calculated
 - **mounting_system:**
 - **fixed:**
 - **slope:**
 - **value:** 0
 - **optimal:** False
 - **azimuth:**
 - **value:** 0
 - **optimal:** False
 - **type:** free-standing
 - **pv_module:**
 - **technology:** c-Si
 - **peak_power:** 540.0
 - **system_loss:** 14.0
 - **economic_data:**
 - **system_cost:** 8.0
 - **interest:** 5.0
 - **lifetime:** 20
- **outputs:**
 - **monthly:**
 - **fixed:**
 - **month:** 1
 - **E_d:** 1159.5
 - **E_m:** 35944.61
 - **H(i)_d:** 2.75
 - **H(i)_m:** 85.15
 - **SD_m:** 3231.77
 - **month:** 2
 - **E_d:** 1486.29
 - **E_m:** 41616.0
 - **H(i)_d:** 3.46

- **H(i)_m:** 96.86
- **SD_m:** 3747.7
- **month:** 3
- **E_d:** 1832.34
- **E_m:** 56802.55
- **H(i)_d:** 4.29
- **H(i)_m:** 132.94
- **SD_m:** 5974.97
- **month:** 4
- **E_d:** 2201.98
- **E_m:** 66059.33
- **H(i)_d:** 5.23
- **H(i)_m:** 156.81
- **SD_m:** 4561.2
- **month:** 5
- **E_d:** 2534.52
- **E_m:** 78570.13
- **H(i)_d:** 6.13
- **H(i)_m:** 189.98
- **SD_m:** 4776.13
- **month:** 6
- **E_d:** 2752.53
- **E_m:** 82575.88
- **H(i)_d:** 6.78
- **H(i)_m:** 203.36
- **SD_m:** 2426.95
- **month:** 7
- **E_d:** 2741.36
- **E_m:** 84982.11
- **H(i)_d:** 6.86
- **H(i)_m:** 212.62
- **SD_m:** 1653.6
- **month:** 8
- **E_d:** 2479.9
- **E_m:** 76876.9
- **H(i)_d:** 6.18
- **H(i)_m:** 191.68
- **SD_m:** 1885.65
- **month:** 9
- **E_d:** 2047.53
- **E_m:** 61425.77
- **H(i)_d:** 5.01
- **H(i)_m:** 150.3
- **SD_m:** 2654.88
- **month:** 10
- **E_d:** 1615.75
- **E_m:** 50088.27

- **H(i)_d:** 3.9
- **H(i)_m:** 120.88
- **SD_m:** 3508.1
- **month:** 11
- **E_d:** 1209.65
- **E_m:** 36289.44
- **H(i)_d:** 2.88
- **H(i)_m:** 86.5
- **SD_m:** 3083.73
- **month:** 12
- **E_d:** 1021.72
- **E_m:** 31673.26
- **H(i)_d:** 2.45
- **H(i)_m:** 75.84
- **SD_m:** 2542.08
- **totals:**
 - **fixed:**
 - **E_d:** 1925.77
 - **E_m:** 58575.35
 - **E_y:** 702904.25
 - **H(i)_d:** 4.67
 - **H(i)_m:** 141.91
 - **H(i)_y:** 1702.91
 - **SD_m:** 1173.11
 - **SD_y:** 14077.29
 - **l_aoi:** −3.2
 - **l_spec:** ?(0)
 - **l_tg:** −8.18
 - **l_total:** −23.56
 - **LCOE_pv:** 0.0
- **meta:**
 - **inputs:**
 - **location:**
 - **description:** Selected location
 - **variables:**
 - **latitude:**
 - **description:** Latitude
 - **units:** decimal degree
 - **longitude:**
 - **description:** Longitude
 - **units:** decimal degree
 - **elevation:**
 - **description:** Elevation
 - **units:** m
 - **meteo_data:**
 - **description:** Sources of meteorological data
 - **variables:**

- ▶ **radiation_db:**
 - **description:** Solar radiation database
- ▶ **meteo_db:**
 - **description:** Database used for meteorological variables other than solar radiation
- ▶ **year_min:**
 - **description:** First year of the calculations
- ▶ **year_max:**
 - **description:** Last year of the calculations
- ▶ **use_horizon:**
 - **description:** Include horizon shadows
- ▶ **horizon_db:**
 - **description:** Source of horizon data
- **mounting_system:**
 - **description:** Mounting system
 - **choices:** fixed, vertical_axis, inclined_axis, two_axis
 - **fields:**
 - ▶ **slope:**
 - **description:** Inclination angle from the horizontal plane
 - **units:** degree
 - ▶ **azimuth:**
 - **description:** Orientation (azimuth) angle of the (fixed) PV system (0 = S, 90 = W, -90 = E)
 - **units:** degree
- **pv_module:**
 - **description:** PV module parameters
 - **variables:**
 - ▶ **technology:**
 - **description:** PV technology
 - ▶ **peak_power:**
 - **description:** Nominal (peak) power of the PV module
 - **units:** kW
 - ▶ **system_loss:**
 - **description:** Sum of system losses
 - **units:** %
- **economic_data:**
 - **description:** Economic inputs
 - **variables:**
 - ▶ **system_cost:**
 - **description:** Total cost of the PV system
 - **units:** user-defined currency
 - ▶ **interest:**
 - **description:** Annual interest
 - **units:** %/y
 - ▶ **lifetime:**
 - **description:** Expected lifetime of the PV system
 - **units:** y

- ▶ **outputs:**
 - **monthly:**
 - **type:** time series
 - **timestamp:** monthly averages
 - **variables:**
 - ▶ **E_d:**
 - **description:** Average daily energy production from the given system
 - **units:** kWh/d
 - ▶ **E_m:**
 - **description:** Average monthly energy production from the given system
 - **units:** kWh/mo
 - ▶ **H(i)_d:**
 - **description:** Average daily sum of global irradiation per square meter received by the modules of the given system
 - **units:** kWh/m2/d
 - ▶ **H(i)_m:**
 - **description:** Average monthly sum of global irradiation per square meter received by the modules of the given system
 - **units:** kWh/m2/mo
 - ▶ **SD_m:**
 - **description:** Standard deviation of the monthly energy production due to year-to-year variation
 - **units:** kWh
 - **totals:**
 - **type:** time series totals
 - **variables:**
 - ▶ **E_d:**
 - **description:** Average daily energy production from the given system
 - **units:** kWh/d
 - ▶ **E_m:**
 - **description:** Average monthly energy production from the given system
 - **units:** kWh/mo
 - ▶ **E_y:**
 - **description:** Average annual energy production from the given system
 - **units:** kWh/y
 - ▶ **H(i)_d:**
 - **description:** Average daily sum of global irradiation per square meter received by the modules of the given system
 - **units:** kWh/m2/d
 - ▶ **H(i)_m:**
 - **description:** Average monthly sum of global irradiation per square meter received by the modules of the given system

- **units:** kWh/m2/mo
- ▶ **H(i)_y:**
 - **description:** Average annual sum of global irradiation per square meter received by the modules of the given system
 - **units:** kWh/m2/y
- ▶ **SD_m:**
 - **description:** Standard deviation of the monthly energy production due to year-to-year variation
 - **units:** kWh
- ▶ **SD_y:**
 - **description:** Standard deviation of the annual energy production due to year-to-year variation
 - **units:** kWh
- ▶ **I_aoi:**
 - **description:** Angle of incidence loss
 - **units:** %
- ▶ **I_spec:**
 - **description:** Spectral loss
 - **units:** %
- ▶ **I_tg:**
 - **description:** Temperature and irradiance loss
 - **units:** %
- ▶ **I_total:**
 - **description:** Total loss
 - **units:** %
- ▶ **LCOE_pv:**
 - **description:** Levelized cost of the PV electricity
 - **units:** system_cost_currency/kWh

9.2. G1

- inputs:
 - location:
 - latitude: 36.66632206941558
 - longitude: -4.460724815726281
 - elevation: 3.0
 - meteo_data:
 - radiation_db: PVGIS-SARAH3
 - meteo_db: ERA5
 - year_min: 2005
 - year_max: 2023
 - use_horizon: True
 - horizon_db: DEM-calculated
 - mounting_system:
 - fixed:
 - slope:
 - value: 0
 - optimal: False
 - azimuth:
 - value: 0
 - optimal: False
 - type: free-standing
 - pv_module:
 - technology: c-Si
 - peak_power: 540.0
 - system_loss: 14.0
 - economic_data:
 - system_cost: 8.0
 - interest: 5.0
 - lifetime: 20
 - outputs:
 - monthly:
 - fixed:
 - month: 1
 - E_d: 1167.93
 - E_m: 36205.98
 - H(i)_d: 2.78
 - H(i)_m: 86.25
 - SD_m: 3249.74
 - month: 2
 - E_d: 1556.97
 - E_m: 43595.26
 - H(i)_d: 3.63
 - H(i)_m: 101.69
 - SD_m: 4119.19

- month: 3
- E_d: 2002.92
- E_m: 62090.59
- H(i)_d: 4.68
- H(i)_m: 145.12
- SD_m: 7152.98
- month: 4
- E_d: 2485.33
- E_m: 74559.87
- H(i)_d: 5.87
- H(i)_m: 176.25
- SD_m: 5660.42
- month: 5
- E_d: 2937.26
- E_m: 91055.19
- H(i)_d: 7.06
- H(i)_m: 218.93
- SD_m: 6147.11
- month: 6
- E_d: 3206.53
- E_m: 96195.99
- H(i)_d: 7.85
- H(i)_m: 235.42
- SD_m: 3149.1
- month: 7
- E_d: 3153.3
- E_m: 97752.36
- H(i)_d: 7.84
- H(i)_m: 243.17
- SD_m: 2020.97
- month: 8
- E_d: 2808.92
- E_m: 87076.64
- H(i)_d: 6.97
- H(i)_m: 216.12
- SD_m: 2562.62
- month: 9
- E_d: 2269.66
- E_m: 68089.66
- H(i)_d: 5.54
- H(i)_m: 166.16
- SD_m: 2976.29
- month: 10
- E_d: 1677.38
- E_m: 51998.79
- H(i)_d: 4.06
- H(i)_m: 125.83

- SD_m: 3784.24
- month: 11
- E_d: 1235.96
- E_m: 37078.93
- H(i)_d: 2.96
- H(i)_m: 88.76
- SD_m: 3196.57
- month: 12
- E_d: 1031.01
- E_m: 31961.28
- H(i)_d: 2.48
- H(i)_m: 76.97
- SD_m: 2579.54
- totals:
 - fixed:
 - E_d: 2130.58
 - E_m: 64805.05
 - E_y: 777660.54
 - H(i)_d: 5.15
 - H(i)_m: 156.72
 - H(i)_y: 1880.65
 - SD_m: 1267.52
 - SD_y: 15210.25
 - l_aoi: -3.51
 - l_spec: ?(0)
 - l_tg: -7.72
 - l_total: -23.42
 - LCOE_pv: 0.0
- meta:
 - inputs:
 - location:
 - description: Selected location
 - variables:
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 - description: Latitude
 - units: decimal degree
 - longitude:
 - description: Longitude
 - units: decimal degree
 - elevation:
 - description: Elevation
 - units: m
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 - description: Sources of meteorological data
 - variables:
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 - description: Solar radiation database

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- **outputs:**
 - **monthly:**

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 - **description:** Average annual energy production from the given system
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 - **H(i)_m:**
 - **description:** Average monthly sum of global irradiation per square meter received by the modules of the given system
 - **units:** kWh/m2/mo
 - **H(i)_y:**

- **description:** Average annual sum of global irradiation per square meter received by the modules of the given system
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 - **description:** Spectral loss
 - **units:** %
- **l_tg:**
 - **description:** Temperature and irradiance loss
 - **units:** %
- **l_total:**
 - **description:** Total loss
 - **units:** %
- **LCOE_pv:**
 - **description:** Levelized cost of the PV electricity
 - **units:** system_cost_currency/kWh

10. Planos

Nota técnica: Todos los materiales deben cumplir con la normativa ISO-800.

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Título del Proyecto	Fecha	Escala
Plano Técnico A3	04/06/2026	